



Foundations of Programming

Assignment No. : 10

PROBLEM STATEMENT:

Write a program in C to simulate a simple calculator that performs basic arithmetic operations such as addition, subtraction, multiplication, and division. The calculator should also perform special operations like computing x^y (power) and $x!$ (factorial).

OBJECTIVE:

- To understand the use of switch-case statements in C.
- To perform basic arithmetic and special mathematical operations.
- To enhance problem-solving skills using menu-driven programs.

THEORY:

A calculator is a basic computing device used to perform arithmetic operations. In C programming, a simple calculator can be implemented using conditional or switch-case statements. Arithmetic operations like addition, subtraction, multiplication, and division are fundamental operations. Special operations such as power (x^y) and factorial ($x!$) require loops for repeated calculations.

PLATFORM:

Linux

ALGORITHM:

Step 1: Start

Step 2: Declare variables

Two numbers x, y , result variable, integer choice, factorial variable $\text{fact} = 1$, and loop counter i .

Step 3: Display menu options:

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Power (x^y)
6. Factorial ($x!$)

Step 4: Read user choice.

Step 5:

- If choice is **1 to 5**, accept two numbers x and y.
- If choice is **6**, accept only one number x.

Step 6: Perform operation using switch case:

- Case 1 $\rightarrow$ result = $x + y$
- Case 2 $\rightarrow$ result = $x - y$
- Case 3 $\rightarrow$ result = $x \times y$
- Case 4 $\rightarrow$
 If $y \neq 0 \rightarrow$ result = x / y
 Else $\rightarrow$ display **division by zero error**
- Case 5 $\rightarrow$ Compute power using loop $\rightarrow$ multiply x by itself y times
- Case 6 $\rightarrow$ Compute factorial using loop $\rightarrow$ fact = $1 \times 2 \times \dots \times x$

Step 7: Display result.

Step 8: Stop.

C PROGRAM:

```
#include <stdio.h>

int main() {

    int choice, i;

    int x, y;

    int power = 1, fact = 1;

    float result;

    printf("\n--- Simple Calculator ---");

    printf("\n1. Addition");

    printf("\n2. Subtraction");

    printf("\n3. Multiplication");

    printf("\n4. Division");

    printf("\n5. Power (x^y)");

    printf("\n6. Factorial (x!)");

    printf("\nEnter your choice: ");

    scanf("%d", &choice);

    switch (choice) {

        case 1:

            printf("Enter two numbers: ");

            scanf("%d %d", &x, &y);

            result = x + y;

            printf("Result = %.2f", result);
```

```
break;
```

case 2:

```
printf("Enter two numbers: ");
```

```
scanf("%d %d", &x, &y);
```

```
result = x - y;
```

```
printf("Result = %.2f", result);
```

```
break;
```

case 3:

```
printf("Enter two numbers: ");
```

```
scanf("%d %d", &x, &y);
```

```
result = x * y;
```

```
printf("Result = %.2f", result);
```

```
break;
```

case 4:

```
printf("Enter two numbers: ");
```

```
scanf("%d %d", &x, &y);
```

```
if (y != 0) {
```

```
    result = (float)x / y;
```

```
    printf("Result = %.2f", result);
```

```
} else {
```

```
    printf("Error! Division by zero.");
```

```
}
```

```
break;
```

case 5:

```
printf("Enter base and exponent: ");
```

```
scanf("%d %d", &x, &y);
```

```
power = 1;
```

```
for (i = 1; i <= y; i++) {
```

```
    power = power * x;
```

```
}
```

```
printf("Result = %d", power);
```

```
break;
```

case 6:

```
printf("Enter a number: ");
```

```
scanf("%d", &x);
```

```
fact = 1;
```

```
for (i = 1; i <= x; i++) {
```

```
    fact = fact * i;
```

```
}
```

```
printf("Factorial = %d", fact);
```

```
break;
```

default:

```
printf("Invalid choice!");
```

```
}
```

```
return 0;
```

}

```
1  #include <stdio.h>
2
3  int main() {
4      int choice, i;
5      int x, y;
6      int power = 1, fact = 1;
7      float result;
8
9      printf("\n--- Simple Calculator ---");
10     printf("\n1. Addition");
11     printf("\n2. Subtraction");
12     printf("\n3. Multiplication");
13     printf("\n4. Division");
14     printf("\n5. Power (x^y)");
15     printf("\n6. Factorial (x!)");
16     printf("\nEnter your choice: ");
17     scanf("%d", &choice);
18
19     switch (choice) {
20
21         case 1:
22             printf("Enter two numbers: ");
23             scanf("%d %d", &x, &y);
24             result = x + y;
25             printf("Result = %.2f", result);
26             break;
27
28         case 2:
29             printf("Enter two numbers: ");
```

```
30     scanf("%d %d", &x, &y);
31     result = x - y;
32     printf("Result = %.2f", result);
33     break;
34
35     case 3:
36         printf("Enter two numbers: ");
37         scanf("%d %d", &x, &y);
38         result = x * y;
39         printf("Result = %.2f", result);
40         break;
41
42     case 4:
43         printf("Enter two numbers: ");
44         scanf("%d %d", &x, &y);
45         if (y != 0) {
46             result = (float)x / y;
47             printf("Result = %.2f", result);
48         } else {
49             printf("Error! Division by zero.");
50         }
51         break;
52
53     case 5:
54         printf("Enter base and exponent: ");
55         scanf("%d %d", &x, &y);
56         power = 1;
```

```

57         for (i = 1; i <= y; i++) {
58             power = power * x;
59         }
60         printf("Result = %d", power);
61         break;
62
63     case 6:
64         printf("Enter a number: ");
65         scanf("%d", &x);
66         fact = 1;
67         for (i = 1; i <= x; i++) {
68             fact = fact * i;
69         }
70         printf("Factorial = %d", fact);
71         break;
72
73     default:
74         printf("Invalid choice!");
75     }
76
77     return 0;
78 }
79

```

INPUT:

User selects an operation from the menu and enters required values.

Example Input:

Choice: 1

Numbers: 10 20

OUTPUT:

```
--- Simple Calculator ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Power (x^y)
6. Factorial (x!)
5. Power (x^y)
5. Power (x^y)
6. Factorial (x!)
Enter your choice: 1
Enter two numbers: 10 20
Result = 30.00
```

CONCLUSION:

Thus, a simple calculator program was successfully implemented using C programming. The program demonstrates the use of switch-case statements, loops, and arithmetic operations.

FAQs:

1. What is a simple calculator program?
2. Why is switch-case preferred in calculator programs?
3. How does the program handle division by zero?
4. How is the power operation (x^y) implemented without using math library functions?
5. What is factorial and how is it calculated using loops in C?